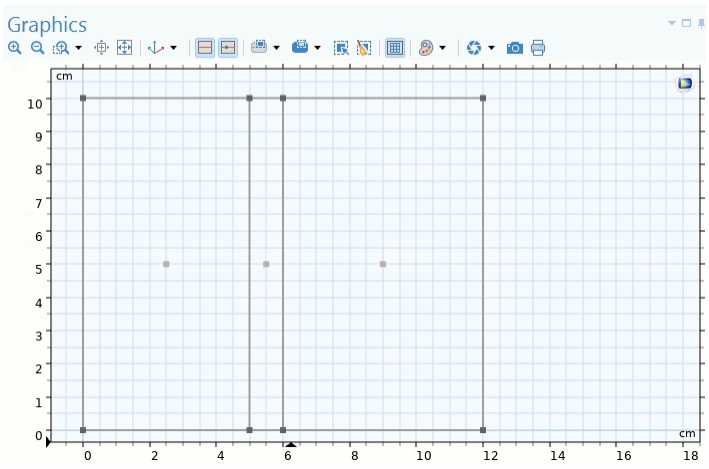
Lab 1: Composite Wall Model

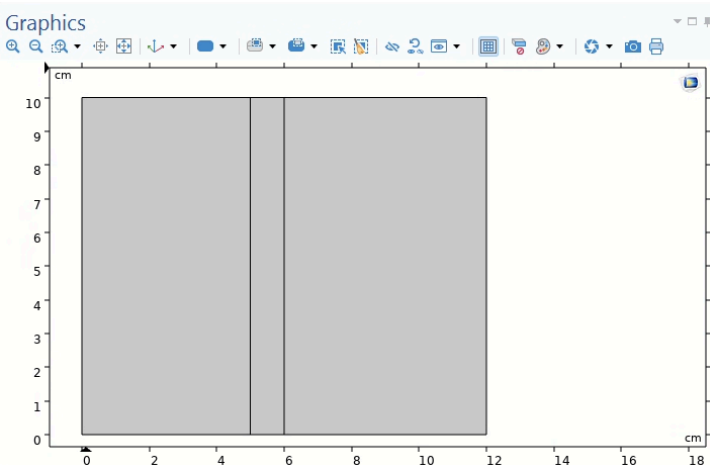
Anne Jumper

SIMULATION PROCESS:

1: Setting up the COMSOL Case

2: Setting up the Geometry

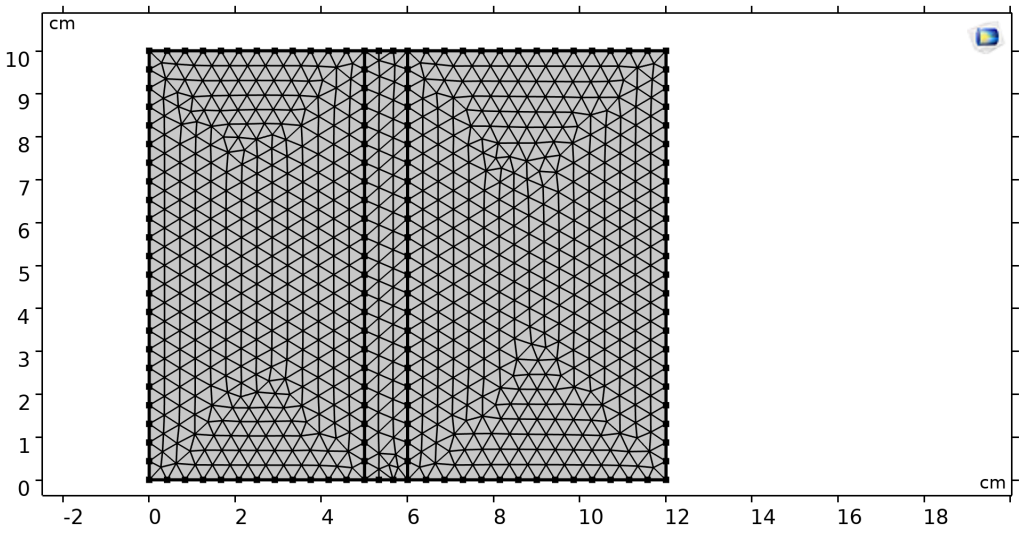




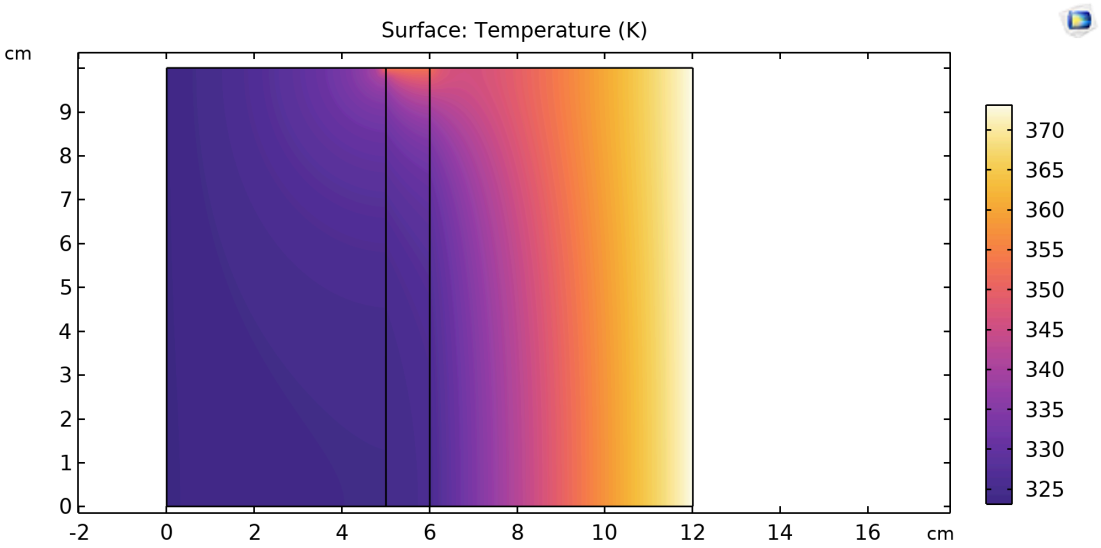
3: Setting the Materials

4: Setting the Boundary Conditions

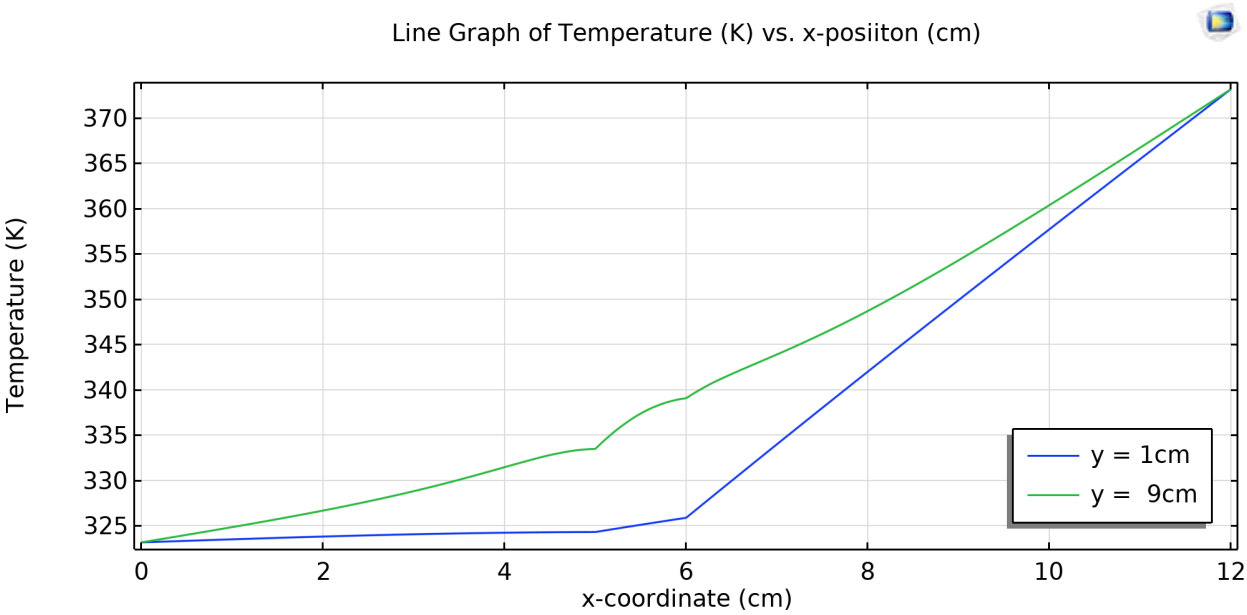
5: Creating a Mesh



6: Computing the Solution

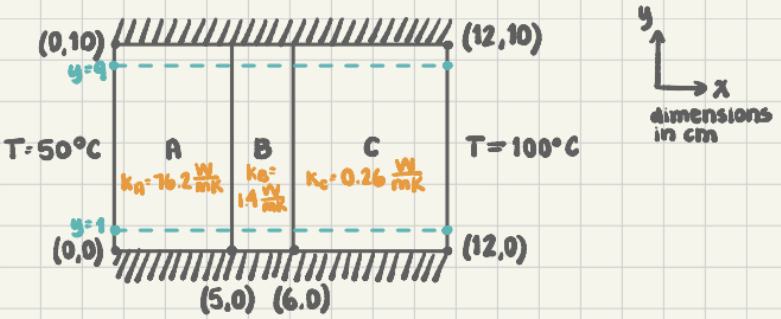


7: Making Plots of the Data



ANALYTICAL SOLUTION

8: 1-Dimensional Analytical Solution



Solve heat equation:

1D Heat Equation with no heat generation: $\frac{d^2T}{dx^2} = 0$

Material A:

$\frac{d^2T}{dx^2} = 0$

$\frac{d}{dx}\left(\frac{dT}{dx}\right) = 0$

$\frac{dT}{dx} = \int 0\,dx$

$\frac{dT}{dx} = A_1$

$\int dT = \int A_1\,dx$

$T_A(x) = A_1 x + A_2$

Material B:

$\frac{d^2T}{dx^2} = 0$

$\frac{d}{dx}\left(\frac{dT}{dx}\right) = 0$

$\frac{dT}{dx} = \int 0\,dx$

$\frac{dT}{dx} = B_1$

$\int dT = \int B_1\,dx$

$T_B(x) = B_1 x + B_2$

Material C:

$\frac{d^2T}{dx^2} = 0$

$\frac{d}{dx}\left(\frac{dT}{dx}\right) = 0$

$\frac{dT}{dx} = \int 0\,dx$

$\frac{dT}{dx} = C_1$

$\int dT = \int C_1\,dx$

$T_C(x) = C_1 x + C_2$

Boundary Conditions:

① Fixed temperature at $x=0$: $T_A(x=0) = 50°C = 323.15K$

② continuous temperature at $x=5$: $T_A(x=5) = T_B(x=5)$

③ continuous heat flux at $x=5$: $k_A A_1 = k_B B_1$

④ continuous temperature at $x=6$: $T_B(x=6) = T_C(x=6)$

⑤ continuous heat flux at $x=$: $k_B B_1 = k_C C_1$

⑥ Fixed temperature at $x=12$: $T_C(x=12) = 100°C = 373.15°C$

6 unknowns! Find using boundary conditions!

Solve for unknowns:

① $T_A(x=0) = A_1(0) + A_2$

$323.15K = A_2$

② $T_A(x=5) = A_1(5cm) + 323.15K$

$T_B(x=5) = B_1(5cm) + B_2$

$5cm \cdot A_1 + 323.15K = 5cm \cdot B_1 + B_2$

$5cm \cdot \left(\frac{1.4}{76.2}\right) B_1 + 323.15K = 5cm \cdot B_1 + B_2$

$B_2 = -4.908cm \cdot B_1 + 323.15K$

$B_2 = -4.908cm \cdot (1.4970 K/cm) + 323.15K$

$B_2 = 315.8024\ K$

③ $k_A A_1 = k_B B_1$

$\left(76.2 \frac{W}{mK}\right) A_1 = \left(1.4 \frac{W}{mK}\right) B_1$

$A_1 = \left(\frac{1.4}{76.2}\right) B_1$

$A_1 = (0.01837)(1.4970 K/cm)$

$A_1 = 0.0275\ K/cm$

④ $T_B(x=6) = B_1(6cm) + B_2$

$T_C(x=6) = C_1(6cm) + C_2$

$6cm \cdot B_1 + B_2 = 6cm \cdot C_1 + C_2$

$6cm \cdot B_1 - 4.908cm \cdot B_1 + 323.15K = 6cm \cdot C_1 + C_2$

$B_1(6 - 4.908cm) + 323.15K = 6cm \cdot C_1 + C_2$

$\left(\frac{0.26}{1.4}\right) C_1 (1.0919cm) + 323.15K = 6cm \cdot C_1 + C_2$

$C_2 = -5.7972cm \cdot C_1 + 323.15K$

$C_2 = -5.7972cm\,(8.0609\ K/cm) + 323.15K$

$C_2 = 276.4191\ K$

⑤ $k_B B_1 = k_C C_1$

$\left(1.4 \frac{W}{mK}\right) B_1 = \left(0.26 \frac{W}{mK}\right) C_1$

$B_1 = \left(\frac{0.26}{1.4}\right) C_1$

$B_1 = (0.1857)(8.0609 K/cm)$

$B_1 = 1.4970\ K/cm$

⑥ $T_C(x=12) = C_1(12cm) + C_2$

$373.15K = 12cm \cdot C_1 + C_2$

$373.15K = 12cm \cdot C_1 - 5.7972cm \cdot C_1 + 323.15K$

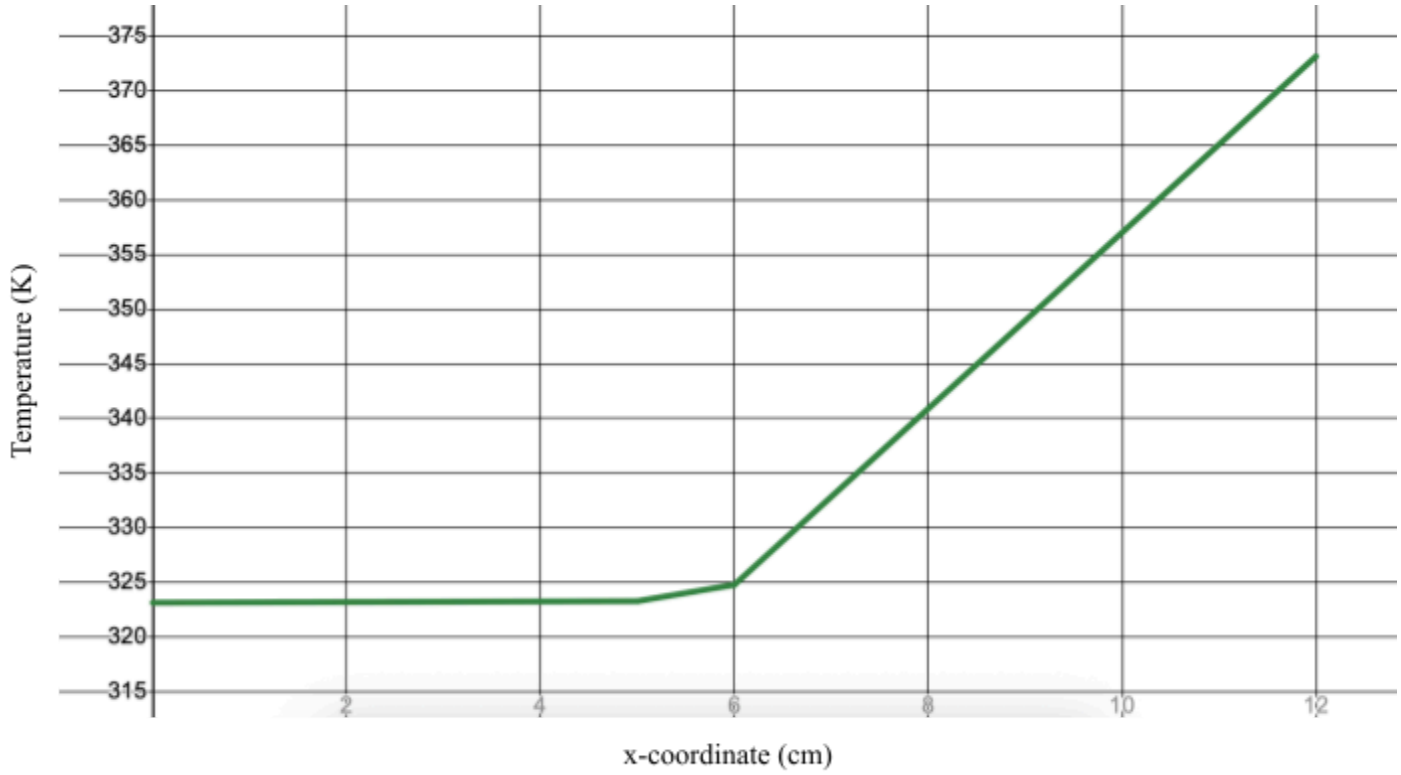
$50K = 6.2028cm \cdot C_1$

$C_1 = 8.0609\ K/cm$

Final Temperature Profile

$$T(x) = \begin{cases} 0.0275x + 323.15 & 0 \le x \le 5cm \\ 1.4970x + 315.8024 & 5 \le x \le 6cm \\ 8.0609x + 276.4191 & 6 \le x \le 12cm \end{cases}$$

Graph of the Analytical Solution



COMPARISON OF RESULTS

9: Results Summary

The analytical solution matches the COMSOL solution most closely when the surface plot is probed along the y = 1cm line. I hypothesize that this is because the COMSOL setup most closely matches the analytical setup along its bottom edge which is fully insulated (unlike the top edge where the glass edge has a fixed 80 degrees celsius temperature). The graphs for both the COMSOL and analytical solutions show linear temperature profiles in all three materials with increasing slopes moving front eh left to the right. However, for the y = 9cm plot, the introduction of the uninsulated face creates a non-linear temperature profile in the glass region, which does not match the linear profile in the analytical results. For y = 1cm however, they both intersect the T axis at around 323 K and about 325 K at the 6 cm distance. Overall, the temperature quantities for both are very similar.